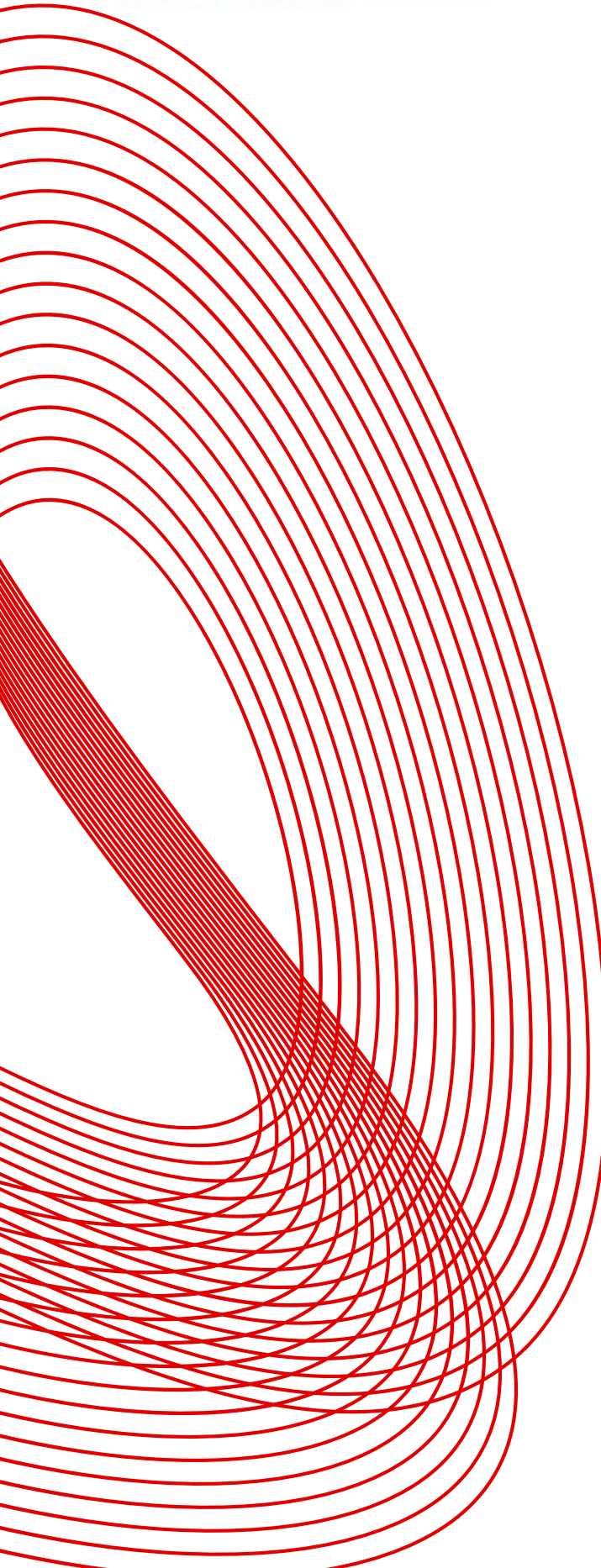


A large, abstract graphic on the left side of the page, composed of numerous concentric, overlapping red lines that form a swirling, tunnel-like shape.

S-IoT Project Report : Drone swarm

Written by :

Emilien PEYON

Noé RAMDANI

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Specifications

Objective

The purpose of this project is to fly a swarm of UAVs (2 to 3 UAVs) and try to lift off a payload using a net.

Functions

- UWB precise location system
- PID Controller for drone stabilization
- UWB Communication for high frequency refreshing data
- Go back and forth keeping the drone formation (following a leader ?)

Deliverables

- Report with all progress, advancements and explanation of technical knowledge
- Functional UWB code
- Methodology and results of calibration of UWB sensors
- Bibliographical Studies
- Video of drone demo
- Functional PID Corrector
- Functional ESP32 code implementation using Multi-threading

Bibliographical Studies

0.1 UWB

0.1.1 Definitions

Ultra-Wideband (UWB) is a technology using radio frequency to exchange data created in 1880. Communication is not the only purpose, it can also be used to locate; it is only possible by using different module, those algorithms and technique will be treated in the next part (0.1.2). In the majority of cases, UWB is deployed for indoor purpose, otherwise, it can be used for outdoor purpose; but it is less shown in scientific papers.

0.1.2 Location algorithm

In order to use UWB as a location sensor. We have to use specific algorithm; in the following subsection, we will explain and show 2 different location algorithms. Firstly, two-way ranging (TWR) and secondly double sided two-way ranging (DS-TWR). Finally, we will describe TDOA, the most advanced algorithms.

Two way ranging (TWR)

The Two-Way Ranging (TWR) algorithm is the easiest and the fastest to implement in a UWB system. In order to explain this algorithm, we will use 2 modules called UWB_1 and UWB_2 .

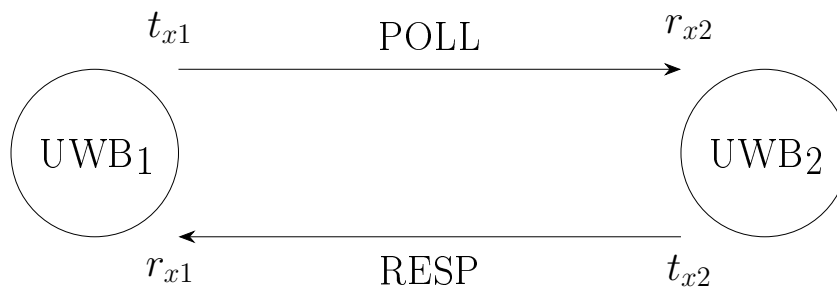


Figure 1: TWR explanation diagram

The algorithm works in 4 steps :

1. UWB_1 sends a **POLL** to UWB_2 . The time of emission t_{x1} is memorized by UWB_1 .
2. UWB_2 receives the **POLL** and memorizes the time of arrival r_{x2} .
3. UWB_2 processes the **POLL**, and precisely plans the time of emission t_{x2} for the **RESP**. It contains (t_{x2}, r_{x2}) .
4. UWB_1 receives the **RESP** and memorizes the r_{x1} .

With all those parameters, we can deduce those variables :

- Total time : $t_{tot} = r_{x1} - t_{x1}$
- Processing time of UWB_2 : $t_{proc} = r_{x2} - t_{x2}$
- Travel time : $t_{travel} = t_{tot} - t_{proc}$

The signal travels back and forth so it did twice the distance (d), and travel speed of a radio signal is equivalent to light speed (c).

We can deduce this equation :

$$\frac{t_{travel} \times c}{2} = \frac{(t_{tot} - t_{proc}) \times c}{2} = \frac{((r_{x1} - t_{x1}) - (r_{x2} - t_{x2})) \times c}{2} \quad (1)$$

We can also determine the induced error of this algorithm.

After calculation (describe the in calculation part page 6), we obtain this equation:

$$\epsilon \approx \frac{t_{proc}(e_a - e_b)}{2} \quad (2)$$

As we can observe, the Time of processing is directly linked to the error. The t_{proc} is around 3,000,000 nanoseconds so, it is inducing a high error.

Double Sided Two way ranging (DS-TWR)

DS-TWR has for main advantage to reduce the error introduced by the t_{proc} . This algorithm also uses UWB_1 and UWB_2 .

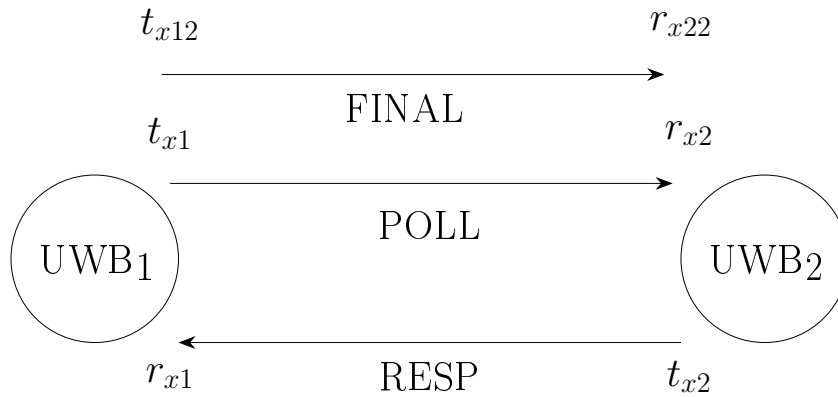


Figure 2: DS-TWR explanation diagram

TDOA

0.2 Measure management

In most cases, data measured by sensors are unclear. It could be a sensor offset, it can also come from crosstalks. Or other, exterior interactions. In order, to obtain clean measurements, and to be able to use them for controlling or any other purpose. We have to develop some filter; in this section, we will describe three filters used for cleaning the UWB signal. Firstly, we will explain the polynomial filter, and secondly, the Kalman filter, then finally the extended Kalman filters

0.2.1 Polynomial Filter

0.2.2 Kalman Filters

0.2.3 Extended Kalman Filter

Calculations

TWR Error calculation :

Let's set:

- True time of flight : T_f
- True time of process : t_{proc}
- Clock drift errors : e_a and e_b

The measured times are:

$$R_a = (2T_f + t_{proc})(1 + e_a) \text{ and } t_{proc_b} = t_{proc}(1 + e_b) \quad (3)$$

The calculated time of flight is:

$$\hat{T}_f = \frac{R_a - t_{proc_b}}{2} \quad (4)$$

By replacing R_a and t_{proc_b} , we obtain :

$$\hat{T}_f = \frac{2T_f(1 + e_a) + t_{proc}(1 + e_a) - t_{proc}(1 + e_b)}{2} \quad (5)$$

By simplifying, we obtain :

$$\hat{T}_f = T_f(1 + e_a) + \frac{t_{proc}(e_a - e_b)}{2} \quad (6)$$

The error is $\hat{T}_f - T_f$, so:

$$Error = T_f \times e_a + \frac{t_{proc}(e_a - e_b)}{2} \quad (7)$$

The time of flight is about 33 nanoseconds, however D is around 3,000,000.
So we can ignore $T_f \times e_a$ cause $T_f \ll t_{proc}$, and we obtain :

$$Error \approx \frac{t_{proc}(e_a - e_b)}{2} \quad (8)$$

DS-TWR Error calculation :